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ON-SCREEN, SPLIT VIRTUAL KEYBOARDS

Abstract

Systems and methods described herein include displaying a keyboard on a display, wherein the keyboard is split into a first keyboard portion and a second keyboard portion; generating first signals pertaining to selection of a character on the first keyboard portion in response to a user interacting with one or more control inputs in a first group of control inputs on a handheld controller; and generating second signals pertaining to selection of a character on the second keyboard portion in response to the user interacting with one or more control inputs in a second group of control inputs on the handheld controller.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is related to U.S. Non-Provisional application Ser. No. _______, filed on the same date as this application, entitled

“ON-SCREEN VIRTUAL KEYBOARDS WITH SIDE-BY-SIDE KEYBOARD PORTIONS,” and identified by Attorney Docket No. 158516 [SIEA23124US00], the entire disclosure of which is hereby fully incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] This invention relates generally to virtual keyboards and, more specifically, to on-screen split virtual keyboards that enable the user to enter characters using two separate cursors.

BACKGROUND

[0003] Typical virtual keyboards generated on screens coupled to computing devices, gaming consoles, and the like generally have a layout resembling the layout of their counterpart physical keyboards. Such virtual keyboards include one on-screen cursor, which may be moved via a controller operated by a user from one letter to the next letter to input each word letter by letter, which is not very efficient, and can be time consuming when entering long words or phrases.

SUMMARY

[0004] In some embodiments, a method includes: displaying a keyboard on a display, wherein the keyboard is split into a first keyboard portion and a second keyboard portion; generating first signals pertaining to selection of a character on the first keyboard portion in response to a user interacting with one or more control inputs in a first group of control inputs on a handheld controller; and generating second signals pertaining to selection of a character on the second keyboard portion in response to the user interacting with one or more control inputs in a second group of control inputs on the handheld controller.

[0005] The first signals may cause a first cursor to move within the first keyboard portion and the second signals may cause a second cursor to move within the second keyboard portion.

[0006] In some aspects, the first signals cause the character in the first keyboard portion to be selected and the second signals cause the character in the second keyboard portion to be selected.

[0007] In certain embodiments, the first group of control inputs are located on a left side of the handheld controller and the second group of control inputs are located on a right side of the handheld controller.

[0008] In one aspect, the user interacting with one or more control inputs in the first group of control inputs on the handheld controller comprises the user interacting with a first analog control input, and the user interacting with one or more control inputs in the second group of control inputs on the handheld controller comprises the user interacting with a second analog control input.

[0009] In certain implementations, the one or more control inputs in the first group of control inputs and the one or more control inputs in the second group of control inputs are configured to allow a first cursor on the first keyboard portion and a second cursor on the second keyboard portion to be moved substantially simultaneously.

[0010] In some embodiments, a non-transitory computer readable storage medium storing one or more computer programs configured to cause a processor-based system to execute steps comprising: displaying a keyboard on a display, wherein the keyboard is split into a first keyboard portion and a second keyboard portion; generating first signals pertaining to selection of a character on the first keyboard portion in response to a user interacting with one or more control inputs in a first group of control inputs on a handheld controller; and generating second signals pertaining to selection of a character on the second keyboard portion in response to the user interacting with one or more control inputs in a second group of control inputs on the controller.

[0011] In certain embodiments, a system comprises: a display; a handheld controller; and a processor-based system in communication with the display and the handheld controller and configured to execute steps comprising: displaying a keyboard on the display, wherein the keyboard is split into a first keyboard portion and a second keyboard portion; generating first signals pertaining to selection of a character on the first keyboard portion in response to a user interacting with one or more control inputs in a first group of control inputs on the handheld controller; and generating second signals pertaining to selection of a character on the second keyboard portion in response to the user interacting with one or more control inputs in a second group of control inputs on the handheld controller.

[0012] A better understanding of the features and advantages of various embodiments of will be obtained by reference to the following detailed description and accompanying drawings which set forth illustrative embodiments of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] Disclosed herein are embodiments of systems and methods pertaining to on-screen split virtual keyboards that enable the user to enter characters using two separate cursors. This description includes drawings, wherein:

[0014] FIG. **1** depicts a display screen view diagram illustrating a prior art virtual keyboard visible thereon;

[0015] FIG. **2** depicts a display screen view diagram illustrating a hand-held controller and a virtual keyboard in accordance with some embodiments;

[0016] FIG. **3** depicts a screen view diagram illustrating the virtual keyboard and the hand-held controller of FIG. **2**, but with the analog joysticks of the hand-held controller being moved to move the corresponding cursors on the virtual keyboard;

[0017] FIG. **4** depicts a display screen view diagram illustrating the virtual keyboard in accordance with some embodiments, where a predictive engine predicts and suggests a word that the user is attempting to type in based on one or more initial letters of the word;

[0018] FIG. **5** is a schematic diagram of a system in accordance with some embodiments;

[0019] FIG. **6** is a block diagram of a computing device in accordance with some embodiments; and

[0020] FIG. **7** is a flow diagram representative of a method in accordance with some embodiments.

[0021] Elements in the figures are illustrated for simplicity and clarity and have not been drawn to scale. For example, the dimensions and/or relative positioning of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present invention. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments of the present invention. Certain actions and/or steps may be described or depicted in a particular order of occurrence while those skilled in the art will understand that such specificity with respect to sequence is not actually required. The terms and expressions used herein have the ordinary technical meaning as is accorded to such terms and expressions by persons skilled in the technical field as set forth above except where different specific meanings have otherwise been set forth herein.

DETAILED DESCRIPTION

[0022] The following description is not to be taken in a limiting sense, but is made merely for the purpose of describing the general principles of exemplary embodiments. Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment.

[0023] Generally speaking, pursuant to various embodiments, systems and methods described herein include displaying a keyboard on a display. The keyboard is split into a first keyboard portion and a second keyboard portion. In response to a user interacting with one or more control inputs in a first group of control inputs on a handheld controller, first signals pertaining to selection of a character on the first keyboard portion are generated. Similarly, in response to the user interacting with one or more control inputs in a second group of control inputs on the handheld controller, second signals pertaining to selection of a character on the second keyboard portion are generated.

[0024] FIG. **1** shows a conventional virtual keyboard **10**, which may be displayed on a screen **11** of a display (e.g., a television, monitor, handheld device, etc.), and which enables a user to type one or more words into a text input field **12** by using a cursor **18** to select one character **20** (e.g., a letter, a number, or a symbol) at a time. The keyboard **10** may include special characters **22** (e.g., up arrow, space bar, back space, etc.), which may be selected with the use of the cursor **18**, or by pressing a specific input **24** (e.g., a button labeled A, B, X, Y, L1, L2, R1, R2, etc.) on a hand-held controller **130** being operated by a user, and the specific input **24** of the controller **130** that may be interacted with by the user to select one of these special characters **22** may be displayed within the virtual keyboard **10** on the screen **11**. Since this keyboard **10** is limited to selection of one character **20** at a time by sequentially moving one cursor **18** to each of the letters of an intended word or phrase, typing in a long word or phrase may become time consuming or tedious for a user.

[0025] FIG. **2** shows a virtual keyboard **100** according to some embodiments described herein, which is displayed on a screen **111** of a display (e.g., a television, monitor, handheld device, etc.), and which facilitates the ease, speed, and accuracy of typing by a user of a hand-held controller **130**. The keyboard **100** includes a first keyboard portion **102** and a second keyboard portion **104**. Each of the first and second keyboard portions **102**, **104** of the exemplary virtual keyboard **100** illustrated in FIG. **2** may include characters **120** such as letters (e.g., a, b, c, etc.), numbers (1, 2, 3, etc.), and special symbols (e.g., question mark, exclamation, coma, period, etc.).

[0026] The virtual keyboard **100** does not have to be limited to only two keyboard portions **102**, **104**. For example, in some embodiments, the virtual keyboard **100** may include four different keyboard portions each including different characters (e.g., letters, numbers, symbols, etc.), and the user is permitted to switch between the first keyboard portion **102** and a third keyboard portion by pressing a pre-determined button input (e.g., L2, etc.) on the hand-held controller **130**, and to switch between the second keyboard portion **102** and a fourth keyboard portion by pressing a pre-determined button input (e.g., R2, etc.) on the hand-held controller **130**. Some other possible alternative configurations of the virtual keyboard **100** are described in more detail in application serial no. ______, filed on the same date as the present application, entitled “ON-SCREEN VIRTUAL KEYBOARDS WITH SIDE-BY-SIDE KEYBOARD PORTIONS,” and identified by Attorney Docket No. 158516 [SIEA23124US00], the entire disclosure of which is hereby fully incorporated by reference herein in its entirety.

[0027] Like the conventional virtual keyboard **10** shown in FIG. **1**, the virtual keyboard **100** according to the embodiment illustrated in FIG. **2** includes an input field **112**, which enables the user to type in letters, numbers, and special symbols/characters by sequentially selecting the appropriate characters **120** on the virtual keyboard **100** using a hand-held controller **130**. Also like the conventional virtual keyboard **10** shown in FIG. **1**, the exemplary virtual keyboard **100** of FIG. **2** includes special characters **122** (e.g., up arrow, space bar, back space, etc.), which may be selected with the use of the cursor **118**, or by pressing a specific input **124** (e.g., button, etc.) on a hand-held controller **130** being operated by a user, causing the specific input **124** that the user interacts with to select one of these special characters **122** to be displayed on the screen **111**.

[0028] Unlike the conventional virtual keyboard **10** of FIG. **1**, in the exemplary keyboard **100** shown in FIG. **2**, the first keyboard portion **102** and the second keyboard portion **104** of the virtual keyboard **100** are visibly separated by a space or a gap **106**, making it clearly visible to a user that the first keyboard portion **102** is separate and distinct from the second keyboard portion **104**. It will be appreciated, however, that, in some embodiments, the first keyboard portion **102** and the second keyboard portion **104** may be separately operable and distinct, but positioned so close together relative to each other such that there would not be a visible gap therebetween.

[0029] Another difference between the exemplary keyboard **100** of FIG. **2** and the conventional keyboard **10** of FIG. **1** is that the keyboard **100** includes not one cursor **18**, but two separate cursors **118** and **119** that facilitate a user's selection of characters **120** on the first and second portions **102**, **104** of the virtual keyboard **100**. In other words, the first keyboard portion **102** includes its own dedicated cursor **118**, which permits the user to select the characters **120** (e.g., letters) of the first keyboard portion **102**, but not the characters **120** (e.g., letters) of the second keyboard portion **104**. Similarly, the second keyboard portion **104** includes its own dedicated cursor **119**, which permits the user to select the characters **120** of the second keyboard portion **104**, but not the characters **120** of the first keyboard portion **102**.

[0030] In the illustrated embodiment, a handheld controller **130** (which will be discussed in more detail below) is used to move the first cursor **118** on the first keyboard portion **102** to select a character **120** desired by the user, and to move the second cursor **119** on the second keyboard portion **104** to select a character **120** desired by the user. In one implementation, the hand-held controller **130** includes a first group of controls **132**, which enables the user to move the first cursor **118** on the first keyboard portion **102** and to select a character **120** desired by the user. This hand-held controller **130** further includes a second group of controls **134**, which enables the user to move the second cursor **119** on the second keyboard portion **104** and to select a character **120** desired by the user.

[0031] In the embodiment illustrated in FIG. **2**, the first group of controls **132** is represented by a first (e.g., left) analog stick of the controller **130** (and may also include a left-hand side directional pad), and the second group of controls **134** is represented by a second (e.g., right) analog stick of the controller **130** (and may also include a right-hand side directional pad). Notably, each of the first and second group of controls **132**, **134** is not limited to an analog stick, and may include only an analog stick, only a directional pad, only a combination of buttons, or a combination thereof.

[0032] In the embodiment illustrated in FIG. **2**, the first keyboard portion **102** of the virtual keyboard **100** is located on the left-hand side of the screen **111** and the second keyboard portion **104** is located on the right-hand side of the screen **111**. To facilitate the logic and ease of use of the virtual keyboard **100** by the user, the first group of controls **132**, which enables the user to move the first cursor **118** on the left-hand side first keyboard portion **102** and to select a character **120** desired by the user, is located on a left-hand side of the hand-held controller **130**. By the same token, the second group of controls **134**, which enables the user to move the second cursor **119** on the right-hand side second keyboard portion **104** and to select a character **120** desired by the user, is located on a right-hand side of the hand-held controller **130**.

[0033] On the exemplary virtual keyboard **100** shown in FIG. **2**, letter “g” is shown as being selected on the first keyboard portion **102** by the first cursor **118**, and the letter “u” is shown as being selected on the second keyboard portion **104** by the second cursor **119**. In the example shown in FIG. **2**, each of the left and right analog sticks **132**, **134** of the hand-held controller **130** is shown as being in a substantially vertical position. In the illustrated embodiment, to move the first cursor **118** of the first keyboard portion **102** from the selected letter “g” to another letter of the first keyboard portion **102**, the user would move the left analog stick **132** of the hand-held controller **130** in the direction of the desired letter. By the same token, to move the second cursor **119** of the second keyboard portion **104** from the selected letter “u” to another letter of the second keyboard portion **104**, the user would move the right analog stick **134** of the hand-held controller **130** in the direction of the desired letter.

[0034] For example, FIG. **3** shows that the directional movement of the left analog stick **132** (by the user) to the left from the substantially vertical orientation of the left analog stick **132** shown in FIG. **2** caused the first cursor **118** to move from the letter “g” to the letter “f,” which is located immediately to the left of the letter “g” on the first keyboard portion **102**. Similarly, FIG. **3** shows that directional movement of the right analog stick **134** (by the user) to the right from the substantially vertical orientation of the right analog stick **134** shown in FIG. **2** caused the second cursor **119** to move from the letter “u” to the letter “i,” which is located immediately to the right of the letter “u” on the second keyboard portion **104**.

[0035] If, for example, the user were to move the left analog stick **132** in an upwardly direction from the substantially vertical orientation shown in FIG. **2**, the first cursor **118** would move from the letter “g” to the letter “t,” which is located immediately above the letter “g” on the first keyboard portion **102**. By the same token, if the user were to move the right analog stick **134** downwardly from the substantially vertical orientation shown in FIG. **2**, the second cursor **119** would move from the letter “u” to the letter “j,” which is located immediately below the letter “u” on the second keyboard portion **104**.

[0036] In the embodiment illustrated in FIG. **2**, each of the first and second cursors **118**, **119** is displayed within the virtual keyboard **100** as a bounding box that surrounds a respective one of the characters **120** (in this case, characters “e” and “h”) of the first and second virtual keyboard portions **102**, **104**. While the bounding box representing each respective cursor **118**, **119** is shown as being rectangular in FIG. **2**, it will be appreciated that the bounding box may be of any other shape. Further, in some embodiments, instead of being represented by a bounding box that surrounds a respective one of the characters (e.g., letters) **120**, the first and second cursors **118**, **119** may be represented within the virtual keyboard **100** as an enlarged version of the respective character on which the respective cursor **118**, **119** is located.

[0037] In other words, in some implementations, if the letter “e” were to be selected by the first cursor **118** on the first keyboard portion **102**, there would be no visible circular bounding box surrounding the letter “e,” but instead the letter “e” would appear in a size that is larger than its original size when not selected by the first cursor **118**. In some other embodiments, instead of being represented by a bounding box that surrounds a respective one of the characters (e.g., letters) **120**, the first and second cursors **118**, **119** may be represented by a color (e.g., a coloring of the letter on which the respective cursor **118**, **119** is located, or a coloring around the letter on which the respective cursor **118**, **119** is located).

[0038] With reference to FIG. **4**, in some embodiments, the virtual keyboard **100** includes word suggestion fields **140**, **142**, **144** that visually indicate a suggested word to the user, which is generated by a prediction engine **180** based on an analysis of the first one or more initial letters of a word that the user starts to type into the input field **112**, a represents a prediction by the prediction engine **180** of the word that the user is attempting to type into the input field **112**. While three word suggestion fields **140**, **142**, **144** are shown in the exemplary keyboard **100** of FIG. **4**, it will be appreciated that the virtual keyboard **100** may include less than three word suggestion fields or more than three word suggestion fields akin to the word suggestion fields **140**, **142**, **144**.

[0039] In the example illustrated in FIG. **4**, the prediction engine **180** has predicted, based on an analysis of the user's selection of the letter “h” in the second keyboard portion **104** using the second cursor **119** (which, after being selected is displayed within the input field **112**), that the user is most likely attempting to type in one of three possible words, namely, “he” (shown in word suggestion field **140**), “her” (shown in word suggestion field **142**), or “his (shown in word suggestion field **144**). In some embodiments, even if only the initial letter (e.g., “h”) appears within the input field **112** of the keyboard **100** after being selected by the user, the user is permitted to complete the full word that the user is attempting to type within the input field **112** by selecting (e.g., via the first or second analog stick **132**, **134** or via another control input of the controller **130**) the full word (e.g., “her”) appearing in one of the word suggestion fields **140**, **142**, **144** that matches the full word that the user is attempting to type into the input field **112**.

[0040] In some embodiments, the predictive engine **180** is programmed with a trained predictive model that is trained to generate, based at least on the complete list of complete words previously typed in by the user into the input field **112**, a prediction of which letter the user is most likely to select next on the virtual keyboard **100** and/or a prediction of the full word that the user is most likely attempting to enter into the input field **112**. In certain aspects, the predictive model of the predictive engine **180** may be continuously updated (e.g., to expand the library of words that may be suggested) over time as the user uses the controller **130** to type in words and/or phrases into the input field **112** of the virtual keyboard **100**. In the example shown in FIG. **4**, the predictive engine **180** has predicted, based at least on the historical data associated with the user, that the user is attempting to enter the word "he," "her," or "his" into the input field **112**.

[0041] Without wishing to be limited to theory, the two-cursor split virtual keyboards **100** described herein enable users to type in words and phrases faster than (e.g., at least 1.5 times faster) and at least as accurately as the existing conventional keyboards, thereby providing a significant time savings for the users and helping the users input their data into their on-screen virtual keyboards via an easy to use and intuitive virtual keyboard design that avoids the slow and tedious data entry offered by conventional on-screen keyboards.

[0042] FIG. **5** shows an exemplary embodiment of a system **200** that enables a user to take advantage of various embodiments of the virtual keyboard **100** described above. The exemplary system **200** shown in FIG. **6** includes a screen **111** (which may be the display of stand-alone device such as a television, monitor, etc., or which may be the display of a portable hand-held device, such as a portable gaming console, smart hone, tablet, laptop, etc.). The screen **111** may display a graphical user interface **160**, which may be a video stream associated with a video game, an on-screen menu, or other visual content which may utilize an on-screen virtual keyboard **100**. The system **200** of FIG. **5** includes a computing device **150** (which may be one or more computing devices as pointed out below) operatively coupled/connected to a display screen **111** and configured to communicate over a network (or connection) **140** with the display screen **111**, one or more hand-held controllers **130** (which may be wired or wireless), and a predictive engine **180** (which may be incorporated into the computing device **150** or stored on a computer/server remote to the computing device **150**).

[0043] The exemplary network **140** depicted in FIG. **5** may be a wide-area network (WAN), a local area network (LAN), a personal area network (PAN), a wireless local area network (WLAN), Wi-Fi, Zigbee, Bluetooth (e.g., Bluetooth Low Energy (BLE) network), or any other internet or intranet network, or combinations of such networks. Generally, communication between various electronic devices of system **200** may take place over hard-wired, wireless, cellular, Wi-Fi or Bluetooth networked components or the like. In some embodiments, one or more electronic devices of system **200** may include cloud-based features or services, such as cloud-based memory storage, cloud-based predictive engine, etc.

[0044] The computing device **150** may be a stationary or portable electronic device, for example, a stationary gaming console, a portable gaming console, a desktop computer, a laptop computer, a tablet, a mobile phone, a single server or a series of communicatively connected servers, or any other electronic device including a control circuit that includes a programmable processor and may be coupled/connected to a display screen **111**. In some embodiments, the computing device **150** is configured for running video games thereon (e.g., from a disc inserted into the computing device **150**, from an onboard memory of the computing device **150**, from a remote server/host, etc.) In some aspects, the computing device **150** is configured for data entry and processing and for communication with other devices of the system **200** via the network **140**.

[0045] With reference to FIG. **6**, an exemplary computing device **150** configured for use with exemplary systems and methods described herein includes a control circuit **310** including a programmable processor (e.g., a microprocessor or a microcontroller) electrically coupled via a connection **315** to a memory **320** and via a connection **325** to a power supply **330**. The control circuit **310** can comprise a fixed-purpose hard-wired platform or can comprise a partially or wholly programmable platform, such as a microcontroller, an application specification integrated circuit, a field programmable gate array, and so on. These architectural options are well known and understood in the art and require no further description here.

[0046] The control circuit **310** can be configured (for example, by using corresponding programming stored in the memory **320** as will be well understood by those skilled in the art) to carry out one or more of the steps, actions, and/or functions described herein. In some embodiments, the memory **320** may be integral to the processor-based control circuit **310** or can be physically discrete (in whole or in part) from the control circuit **310** and is configured non-transitorily store the computer instructions that, when executed by the control circuit **310**, cause the control circuit **310** to behave as described herein. (As used herein, this reference to “non-transitorily” will be understood to refer to a non-ephemeral state for the stored contents (and hence excludes when the stored contents merely constitute signals or waves) rather than volatility of the storage media itself and hence includes both non-volatile memory (such as read-only memory (ROM)) as well as volatile memory (such as an erasable programmable read-only memory (EPROM))). Accordingly, the memory and/or the control unit may be referred to as a non-transitory medium or non-transitory computer readable medium.
[0047] The control circuit **310** of the computing device **150** is also electrically coupled via a connection **335** to an input/output **340** that can receive signals from, for example, from one or more hand-held controllers **130**, predictive engine **180**, etc. The input/output **340** of the computing device **150** can also send signals to other devices, for example, a signal to the electronic display **111**, predictive engine **180**, etc.
[0048] The processor-based control circuit **310** of the computing device **150** shown in FIG. **6** is electrically coupled via a connection **345** to a user interface **350**, which may include inputs **370** (e.g., buttons, ports, touch screens, etc.) that permit an operator of the computing device **150** to manually control the computing device **150** by inputting commands via touch button operation and/or voice commands and/or via a physically connected device (e.g., hand-held controller **130**, etc.). Possible commands may, for example, cause the computing device **150** to turn on and off, reset, eject a video game disc, etc. In some embodiments, the user interface **350** of the computing device **150** may also include a speaker **360** that provides audible feedback (e.g., notifications, alerts, etc.) to the operator of the computing device **150**. It will be appreciated that the performance of such functions by the processor-based control circuit **310** of the computing device **150** is not dependent on a human operator, and that the control circuit **310** of the computing device **150** may be programmed to perform such functions without a human operator.
[0049] In the embodiment illustrated in FIG. **5**, the system **200** includes a predictive engine **180** that is configured to obtain (e.g., from the computing device **150** and over the network **140**) and process (e.g., via an artificial intelligence/machine learning model) the data representative of the text input (e.g., words and phrases) entered by the user via the hand-held controller **130** into the input field **112** of the virtual keyboard **100**. This processing of the user's input of letters or numerical characters permits the predictive engine **180** to generate one or more predictions of the words that the user is attempting to type into the virtual keyboard **100**.
[0050] In the embodiment illustrated in FIG. **5**, the predictive engine **180** and the computing device **150** are shown as being implemented as two separate physical devices (which may be located at the same physical location or in different physical locations). It will be appreciated, however, that the computing device **150** and the predictive engine **180** may be implemented as a single physical device. In some aspects, the predictive engine **180** may be stored, for example, on non-volatile storage media (e.g., a hard drive, flash drive, or removable optical disk) internal or external to the computing device **150**, or internal or external to computing devices separate and distinct from the computing device **150**. In some embodiments, the predictive engine **180** may be cloud-based.
[0051] In certain implementations, the predictive engine **180** processes the data representing the text being input by the user into the input field **112** of the virtual keyboard **100** by executing one or more trained machine learning modules and/or trained neural network modules/models. In certain aspects, the neural network executed by the predictive engine **180** (by itself or via the control circuit **310** of the computing device **150**) may be a deep convolutional neural network. The neural

network module/model may be trained using various data sets, including, but not limited to: letter-by-letter sequential entries made by the user when typing any word or character into the virtual keyboard **100**, a library of complete words previously entered by the user into the virtual keyboard **100**, a dictionary-like library of possible words that may be suggested to the user, etc.

[0052] In some embodiments, the predictive engine **180** may be trained to analyze the user's text input into the input field **112** of the virtual keyboard **100** using one or more machine learning algorithms, including but not limited to Linear Regression, Logistic Regression, Decision Tree, SVM, Naïve Bayes, kNN, K-Means, Random Forest, Dimensionality Reduction Algorithms, and Gradient Boosting Algorithms. In some embodiments, the trained machine learning/neural network module/model of the predictive engine **180** includes a computer program code stored in a memory and/or executed by a control circuit (e.g., the control circuit **310**) to process an in-progress text input by the user to generate a list of predicted words that the user is attempting to type into the virtual keyboard.

[0053] As noted above, in some implementations, the predictive engine **180** is programmed with a trained machine learning model that is trained to generate, based on an available list of the full words previously typed in by the user into the input field **112**, a prediction of which letter the user is most likely to select next on the virtual keyboard **100** and/or a prediction of the full word that the user is most likely attempting to enter into the input field **112**. In certain aspects, the predictive model of the predictive engine **180** may be continuously updated in real time as a result of the user typing in new (i.e., previously unentered) words into the virtual keyboard **100**.

[0054] The exemplary system **200** shown in FIG. **5** further includes a hand-held controller **130**. Generally, gaming-specific computing devices/entertainment systems such as stationary or portable gaming consoles (e.g., Sony PlayStation, PlayStation Portable, Microsoft X Box, Nintendo Switch, etc.) include a hand-held game controller, which permits a user to enter in-game or menu commands or other instructions into to the computing/gaming system to control a video game, a gaming-associated stream, or a gaming-associated menu.

[0055] The exemplary hand-held controller **130** illustrated in FIGS. **2-3** includes various control inputs. For example, the band-held controller **130** includes a first analog joystick **132** and a second analog joystick **134**, which may be referred to herein as a “first group of control inputs” and a “second group of control inputs,” respectively. In some embodiments, a manipulated variable of a first or second analog stick **132**, **134** (e.g., the directional movement caused by a user using the user's finger(s)) is converted from an analog value into a digital value, which is transmitted by the hand-held controller **130** to the computing device **150**, in turn causing a responsive in-game action, which is visible to the user on the screen **111**.

[0056] In some embodiments, the hand-held controller **130** is provided with various button inputs **124** that may be pushed by a user. In the exemplary embodiment shown in FIG. **2**, the controller **130** includes a third button (e.g., L2, etc.) that permits the user to switch the letters of the first and second virtual keyboard portions between lower case letters and capital letters; a fourth button (e.g., triangle, etc.) that permits the user to create a space between the letters, a fifth button (e.g., R3, etc.) that permits the user to select a special character that is not a letter; a sixth button (e.g., square, etc.) that permits the user to undo the selectin of one or more of the letters; a seventh button (e.g., L1, etc.) that permits the user to enter a letter selected by the first cursor **118** to be entered into the text input field **112**; an eighth button (e.g., R1, etc.) that permits the user to enter a letter selected by the second cursor **119** to be entered into the text input field **112**; and a ninth button (e.g., touch pad, etc.) that may be touched or pressed by the user to, for example, move the first and second cursors **118**, **119** or select letters via the first and second cursors **118**, **119**. It will be appreciated that the number of button inputs **124** of the hand-held controller **130** and the functions assigned to each of the button inputs **124** of the hand-held controller **130** are shown/described by way of example only.

[0057] In some embodiments, after a user uses the first analog stick **132** to move the first cursor

118 to the user-desired letter of the first keyboard portion **102**, the user then presses one of the button inputs **124** (e.g., L1) of the hand-held controller (which may be located on a left-hand side of the hand-held controller **130** just like the first analog stick **132**) to enter a selection of the user-desired letter and cause the user-selected letter to appear in the input field **112** of the virtual keyboard **100**. Similarly, after a user uses the second analog stick **134** to move the second cursor **119** to the user-desired letter of the second keyboard portion **104**, the user presses one of the button inputs **124** (e.g., R1) of the hand-held controller **130** (which may be located on a right-hand side of the hand-held controller **130** just like the second analog stick **134**) to enter a selection of the user-desired letter and cause this letter to appear in the input field **112** of the virtual keyboard **100**.

[0058] Notably, the exemplary virtual keyboard **100** illustrated in FIG. **2** includes a character field **122**, which visually indicates both the function (e.g., space bar) of the character field **122** (in this case, by a bracket representing a space bar) and the button (in this case, the triangle button input **124**) of the controller **130** that executes this particular function. In some embodiments, the first and second analog sticks **132**, **134** may themselves be pressable, and the controller **130** may be configured such that, after the user navigates the first and second cursors **118**, **119** via their respective first and second analog sticks **132**, **134** to the user-desired letter, instead of pressing a separate button input **124** (e.g., L1 or R1) of the controller **130**, the user may press the first analog stick **132** to select the letter underlying the first cursor **118** and the second analog stick **134** to select the letter underlying the second cursor **119**.

[0059] In certain embodiments, the first and second analog sticks **132**, **134** are configured to have user-adjustable dead zones, i.e., zones, where directional movement of an analog stick does not cause a responsive action on the display screen **111**. In other words, the dead zone of each of the first and second analog sticks **132**, **134** may be defined as an area (or an imaginary perimeter) around an analog stick **132**, **134**, where movement of the analog stick **132**, **134** does not input a command into the controller **130** until the analog stick **132**, **134** is moved by the user out of the dead zone.

[0060] The dead zone of a directional analog stick of a hand-held controller **130** may be expressed in degrees. For example, if the dead zone of the first analog stick **132** is 20 degrees, movement of the first analog stick **132** up to 20 degrees relative to the vertical would not cause the first cursor **118** to move on the first keyboard portion **102**. On the other hand, the movement of the first analog stick **132** by 21 or more degrees relative to the vertical would cause the first cursor **118** to move on the first keyboard portion **102** in a direction corresponding to the movement of the direction of movement of the first analog stick **132**.

[0061] In certain embodiments, the hand-held controller **130** is configured such that, when the user holds the first analog stick **132** outside of its respective dead zone, a first delay timer is triggered to delay the movement of the first cursor **118** between adjacent ones of the letters of the first keyboard portion **102** of the virtual keyboard **100** for a predetermined period of time. Similarly, in some implementations, the hand-held controller **130** is configured such that, when the user holds the second analog stick **132** outside of its respective dead zone, a second delay timer is triggered to delay the movement of the second cursor **119** between adjacent ones of the letters of the second keyboard portion **104** of the virtual keyboard **100** for a predetermined period of time.

[0062] The delay timers associated with the first and second analog sticks **132**, **134** may be set independently from one another and may be identical or different from one another. In one embodiment, the hand-held controller **130** may be configured such that the length of the first and second delay timers decreases in proportion to an increasing number of adjacent letters to be moved across by the first and second cursors **118**, **119** in response to a directional tilt of the first and second analog sticks **132**, **134** relative to the vertical. The aim of this feature is to prevent the cursors **118**, **119** from moving too fast across adjacent letters of their respective first and second keyboard portions **102**, **104**, thereby reducing the chances that the directional movement of the first and second analog sticks **132**, **134** is too fast to permit the user to stop the first and second cursor

118, **119** on a desired letter instead of going past it.

TABLE-US-00001 TABLE 1 Possible Hand-Held Controller Delay Timer Scheme Time Passed Delay of Next Key (Letter) Keys (Letters) (milliseconds) Movement (milliseconds) Moved 0 120 1 120 35 2 155 29 3 185 26 4 210 23 5 233 21 6 254 20 7

[0063] Table 1 above shows a possible delay timer scheme that may be programmed into the hand-held controller **130** to facilitate precise movements of the first and second cursors **118**, **119** via the first and second analog sticks **132**, **134**. In some aspects, when the user holds an analog stick of the controller **130** outside of its respective dead zone, a delay function gets run which prevents key (i.e., letter) navigation from being too fast, but if the user returns the analog stick of the controller **130** to the dead zone, the delay function gets reset and the key movement occurs. This allows the user to make precise movements between two adjacent letters with the first and second analog sticks **132**, **134** while pressing the first and second analog sticks **132**, **134** or while holding down the first and second analog sticks **132**, **134** to slide across adjacent letters.

[0064] FIG. **7** shows an exemplary embodiment of a method **400** of providing a virtual keyboard **100**. The method **400** includes displaying a virtual keyboard **100** on a display **111** such that the virtual keyboard **100** is split into a first keyboard portion **102** and a second keyboard portion **104** (step **410**). As discussed above, in some embodiments, the display **111** may be a stand-alone display such as a television, a monitor, etc. In other embodiments, the display **111** may be incorporated into a computing device **150**, which may include but is not limited to a hand-held gaming console, a tablet, a laptop, a mobile phone, etc.

[0065] The exemplary illustrated method **400** further includes generating first signals pertaining to selection of a character **120** on the first keyboard portion **102** in response to a user interacting with one or more control inputs in a first group of control inputs (e.g., a first analog stick **132**) on a handheld controller **130** (step **420**). In addition, the method **400** includes generating second signals pertaining to selection of a character **120** on the second keyboard portion **104** in response to the user interacting with one or more control inputs in a second group of control inputs (e.g., a second analog stick **134**) on the handheld controller **130** (step **430**).

[0066] In some embodiments, one or more of the embodiments, methods, approaches, schemes, and/or techniques described above may be implemented in one or more computer programs or software applications executable by a processor-based apparatus or system. By way of example, such processor-based system may comprise a smartphone, tablet computer, virtual reality (VR), augmented reality (AR), or mixed reality (MR) system, entertainment system, game console, mobile device, computer, workstation, gaming computer, desktop computer, notebook computer, server, graphics workstation, client, portable device, pad-like device, communications device or equipment, etc. Such computer program(s) or software may be used for executing various steps and/or features of the above-described methods, schemes, and/or techniques. That is, the computer program(s) or software may be adapted or configured to cause or configure a processor-based apparatus or system to execute and achieve the functions described herein. For example, such computer program(s) or software may be used for implementing any embodiment of the above-described methods, steps, techniques, schemes, or features. As another example, such computer program(s) or software may be used for implementing any type of tool or similar utility that uses any one or more of the above-described embodiments, methods, approaches, schemes, and/or techniques. In some embodiments, one or more such computer programs or software may comprise a VR, AR, or MR application, communications application, object positional tracking application, a tool, utility, application, computer simulation, computer game, video game, role-playing game (RPG), other computer simulation, or system software such as an operating system, BIOS, macro, or other utility. In some embodiments, program code macros, modules, loops, subroutines, calls, etc., within or without the computer program(s) may be used for executing various steps and/or features of the above-described methods, schemes and/or techniques. In some embodiments, such computer program(s) or software may be stored or embodied in a non-transitory computer readable storage or recording medium or media, such as a tangible computer readable storage or recording medium or media. In some embodiments, such computer program(s) or software may be stored or embodied in transitory computer readable storage or recording medium or media, such as in one or more transitory forms of signal transmission (for example, a propagating electrical or electromagnetic signal).

[0067] Therefore, in some embodiments the present invention provides a computer program product comprising a medium for embodying a computer program for input to a computer and a computer program embodied in the medium for causing the computer to perform or execute steps comprising any one or more of the steps involved in any one or more of the embodiments, methods, approaches, schemes, and/or techniques described herein. For example, in some embodiments the present invention provides one or more non-transitory computer readable storage mediums storing one or more computer programs adapted or configured to cause a processor-based apparatus or system to execute steps comprising any one or more of the embodiments, methods, approaches, schemes, and/or techniques described herein.

[0068] While the invention herein disclosed has been described by means of specific embodiments and applications thereof, numerous modifications and variations could be made thereto by those skilled in the art without departing from the scope of the invention set forth in the claims.

Claims

1. A method comprising: displaying a keyboard on a display, wherein the keyboard is split into a first keyboard portion and a second keyboard portion; generating first signals pertaining to selection of a character on the first keyboard portion in response to a user interacting with one or more control inputs in a first group of control inputs on a handheld controller; and generating second signals pertaining to selection of a character on the second keyboard portion in response to the user interacting with one or more control inputs in a second group of control inputs on the handheld controller.

2. The method of claim 1, wherein: the first signals cause a first cursor to move within the first keyboard portion; and the second signals cause a second cursor to move within the second keyboard portion.

3. The method of claim 1, wherein: the first signals cause the character in the first keyboard portion to be selected; and the second signals cause the character in the second keyboard portion to be selected.

4. The method of claim 1, wherein: the first group of control inputs are located on a left side of the handheld controller; and the second group of control inputs are located on a right side of the handheld controller.

5. The method of claim 1, wherein: the user interacting with one or more control inputs in the first group of control inputs on the handheld controller comprises the user interacting with a first analog control input; and the user interacting with one or more control inputs in the second group of control inputs on the handheld controller comprises the user interacting with a second analog control input.

6. The method of claim 1, wherein the one or more control inputs in the first group of control inputs and the one or more control inputs in the second group of control inputs are configured to allow a first cursor on the first keyboard portion and a second cursor on the second keyboard portion to be moved substantially simultaneously.

7. A non-transitory computer readable storage medium storing one or more computer programs configured to cause a processor-based system to execute steps comprising: displaying a keyboard on a display, wherein the keyboard is split into a first keyboard portion and a second keyboard portion; generating first signals pertaining to selection of a character on the first keyboard portion in response to a user interacting with one or more control inputs in a first group of control inputs on a handheld controller; and generating second signals pertaining to selection of a character on the second keyboard portion in response to the user interacting with one or more control inputs in a second group of control inputs on the handheld controller.

8. The non-transitory computer readable storage medium of claim 7, wherein: the first signals cause a first cursor to move within the first keyboard portion; and the second signals cause a second cursor to move within the second keyboard portion.

9. The non-transitory computer readable storage medium of claim 7, wherein: the first signals cause the character in the first keyboard portion to be selected; and the second signals cause the character in the second keyboard portion to be selected.

10. The non-transitory computer readable storage medium of claim 7, wherein: the first group of control inputs are located on a left side of the handheld controller; and the second group of control inputs are located on a right side of the handheld controller.

11. The non-transitory computer readable storage medium of claim 7, wherein: the user interacting with one or more control inputs in the first group of control inputs on the handheld controller comprises the user interacting with a first analog control input; and the user interacting with one or more control inputs in the second group of control inputs on the handheld controller comprises the user interacting with a second analog control input.

12. The non-transitory computer readable storage medium of claim 7, wherein the one or more control inputs in the first group of control inputs and the one or more control inputs in the second group of control inputs are configured to allow a first cursor on the first keyboard portion and a second cursor on the second keyboard portion to be moved substantially simultaneously.

13. A system, comprising: a display; a handheld controller; and a processor-based system in communication with the display and the handheld controller and configured to execute steps comprising, displaying a keyboard on the display, wherein the keyboard is split into a first keyboard portion and a second keyboard portion; generating first signals pertaining to selection of a character on the first keyboard portion in response to a user interacting with one or more control inputs in a first group of control inputs on the handheld controller; and generating second signals pertaining to selection of a character on the second keyboard portion in response to the user interacting with one or more control inputs in a second group of control inputs on the handheld controller.

14. The system of claim 13, wherein: the first signals cause a first cursor to move within the first keyboard portion; and the second signals cause a second cursor to move within the second keyboard portion.

15. The system of claim 13, wherein: the first signals cause the character in the first keyboard portion to be selected; and the second signals cause the character in the second keyboard portion to be selected.

16. The system of claim 13, wherein: the first group of control inputs are located on a left side of the handheld controller; and the second group of control inputs are located on a right side of the handheld controller.

17. The system of claim 13, wherein: the user interacting with one or more control inputs in the first group of control inputs on the handheld controller comprises the user interacting with a first analog control input; and the user interacting with one or more control inputs in the second group of control inputs on the handheld controller comprises the user interacting with a second analog control input.

18. The system of claim 13, wherein the one or more control inputs in the first group of control inputs and the one or more control inputs in the second group of control inputs are configured to allow a first cursor on the first keyboard portion and a second cursor on the second keyboard portion to be moved substantially simultaneously.